ADVANCED EXCEL MINI PROJECT

Project Documentation

Retail Store Sales Analysis

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Platform: Agileology (EdTech & Upskilling Platform)

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Duration: Mar 15, 2026 – Mar 18, 2026

DATASET SNAPSHOT			
Total Records	Columns	Date Range	Unique Customers
12,575 rows	**11 original**	**2022 - 2025**	**25 customers**

TOOLS & PLATFORM
MS Excel (Advanced) | Agileology Platform | GitHub

GitHub Repository: https://github.com/swaroop456/Advanced-Excel-Retail-Sales-Analysis

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1. PROJECT OVERVIEW & OBJECTIVES

This project was completed as part of the Advanced Excel Mini Project assigned by Agileology, an EdTech and Upskilling Platform. The objective was to perform complete end-to-end data analysis on a Retail Store Sales dataset using Microsoft Excel (Advanced) — covering data cleaning, transformation, advanced formulas, pivot tables, charts and an interactive KPI dashboard.

OBJECTIVE	DESCRIPTION
Data Cleaning	Clean raw data — fix blanks, check duplicates, standardize formats
Advanced Formulas	Apply 17+ formulas: SUMIFS, XLOOKUP, Dynamic Arrays, Nested IF
Pivot Tables	Build 4 Pivot Tables with calculated fields, slicers and timeline
Advanced Charts	Create Combo, Sparkline and Waterfall charts for deeper insights
KPI Dashboard	Design interactive dashboard with linked KPIs, charts and filters
Documentation	Document all steps, formulas and findings inside the workbook

2. DATASET DESCRIPTION

COLUMN NAME	DATA TYPE	DESCRIPTION
Transaction ID	Text	Unique identifier for each transaction
Customer ID	Text	Unique identifier for each customer
Category	Text	Product category (8 categories)
Item	Text	Specific item purchased
Price Per Unit	Number	Unit price of the item purchased
Quantity	Number	Number of units purchased
Total Spent	Number	Total transaction value
Payment Method	Text	Cash / Credit Card / Digital Wallet
Location	Text	Online or In-store transaction
Transaction Date	Date	Date of transaction (Jan 2022 to Jan 2025)
Discount Applied	Boolean	TRUE if discount was applied / FALSE if not

3. DATA CLEANING & TRANSFORMATION

STEP	ACTION	COLUMN AFFECTED	RECORDS FIXED
Step 1	Protected original data by copying to Cleaned Data sheet	All columns	—
Step 2	Converted data range to Excel Table named Sales Data (Ctrl+T)	Entire dataset	—
Step 3	Checked and removed duplicates — zero duplicates found	All columns	0
Step 4	Fixed blank cells with FALSE (Boolean value)	Discount Applied	4,199
Step 5	Fixed blank cells with "Unknown"	Item	1,213
Step 6	Fixed blank cells with 0	Price Per Unit	609
Step 7	Fixed blank cells with 0	Quantity & Total Spent	604
Step 8	Added helper column: Month name extracted from date	Column L (Month)	12,575
Step 9	Added helper column: Year extracted from date	Column M (Year)	12,575
Step 10	Added helper column: Month-Year combined label	Column N (Month-Year)	12,575

Step 11	Added Sales Tier using Nested IF: High/Medium/Low	Column O (Sales Tier)	12,575

4. ADVANCED FORMULAS & FUNCTIONS

FORMULA	CATEGORY	PURPOSE & RESULT
=SUM(SalesData[Total Spent])	Aggregate	Total revenue from all transactions — Result: $1,552,071
=AVERAGE(SalesData[Total Spent])	Aggregate	Average order value — Result: $129.65
=COUNTA(SalesData[Transaction ID])	Aggregate	Total transactions count — Result: 12,575
=COUNTA(UNIQUE(SalesData[Customer ID]))	Dynamic Array	Total unique customers — Result: 25
=COUNTIF(SalesData[Discount Applied],TRUE)	Conditional	Orders with discount applied — Result: 4,219
=SUMIF(SalesData[Category],"Butchers",SalesData[Total Spent])	Conditional	Total sales for one category — Result: $208,118
=COUNTIF(SalesData[Category],"Beverages")	Conditional	Count of orders per category
=SUMIFS(SalesData[Total Spent],SalesData[Location],"Online",SalesData[Year],2024)	Multi-condition	Online sales for 2024 — Result: $265,246.50
=XLOOKUP(B51,SalesData[Transaction ID],SalesData[Total Spent],"Not Found")	Lookup	Look up transaction details by ID
=UNIQUE(SalesData[Category])	Dynamic Array	Lists all unique categories dynamically
=SORT(UNIQUE(SalesData[Category]))	Dynamic Array	Sorted unique categories alphabetically
=FILTER(SalesData[Item],SalesData[Total Spent]>500)	Dynamic Array	Returns only high value transactions
=LARGE(SalesData[Total Spent],{1,2,3,4,5})	Dynamic Array	Returns top 5 highest sales values
=IF([@[Total Spent]]>500,"High",IF([@[Total Spent]]>200,"Medium","Low"))	Logical	Classifies each transaction: High/Medium/Low
=TEXT([@[Transaction Date]],"MMMM")	Date/Text	Extracts full month name from date
=YEAR([@[Transaction Date]])	Date/Text	Extracts year number from date
=TEXT([@[Transaction Date]],"MMM-YYYY")	Date/Text	Creates combined Month-Year label

5. PIVOT TABLES & PIVOT CHARTS

PIVOT TABLE	ROWS	VALUES	FILTER	CHART TYPE
PT1 — Sales by Category	Category	SUM of Total Spent + Calculated Field (Avg Sale Value)	Year	Clustered Column
PT2 — Monthly Sales Trend	Month-Year	SUM of Total Spent	Category	Line Chart
PT3 — Payment Method	Payment Method	SUM of Total Spent	Location	Pie Chart
PT4 — Online vs In-Store	Location	COUNT of Transaction ID	Year	Column Chart

All 4 Pivot Tables are connected to Category, Location and Payment Method Slicers and a Transaction Date Timeline for interactive filtering.

6. ADVANCED CHARTS

CHART TYPE	DATA USED	PURPOSE
Combo Chart	Category vs Total Sales + Order Count	Compare sales value and order volume side by side using dual axes
Sparklines	Monthly Total Spent values	Show mini trend lines inline within the Analysis sheet cells
Waterfall Chart	Month-Year vs Total Spent from PT2	Visualize monthly revenue gains and losses with cumulative flow

7. KPI DASHBOARD

An interactive KPI Dashboard was built on a dedicated Dashboard sheet, bringing together all key metrics, charts and filters in one view. The dashboard includes:

- 6 KPI summary cards — Total Revenue, Avg Order Value, Total Transactions, Total Customers, Discounted Orders, Non-Discounted Orders
- 3 Pivot Charts — Sales by Category (Column), Monthly Sales Trend (Line), Payment Method Share (Pie)
- 3 Interactive Slicers — Category, Location, Payment Method — all connected to all 4 Pivot Tables
- 1 Transaction Date Timeline — for filtering by month/year across all pivots
- Professional color theme with borders and formatted KPI cards

8. DATA ANALYSIS TOOLS

TOOL	HOW IT WAS USED

Goal Seek	Used to determine what Total Spent value would be needed to reach a target revenue of $200,000 — via Data > What-If Analysis > Goal Seek
Scenario Manager	Created 3 scenarios (Low Sales: $50, Base Sales: $200, High Sales: $500) to model different sales performance outcomes

9. KEY FINDINGS & INSIGHTS

$1,552,071	12,575	25	$129.65
Total Revenue (2022-2025)	Total Transactions	Unique Customers	Avg Order Value

1

Highest Selling Category

Butchers | $208,118

Butchers generated the highest total revenue at $208,118, followed by Electric Household Essentials at $203,813.50. Milk Products had the lowest sales at $180,112.

2

Top Payment Method

Cash | $537,710

Cash was the most popular payment method generating $537,710 in total sales. Digital Wallet ($507,279) and Credit Card ($507,082) were nearly equal — showing all 3 methods are widely used.

3

Online vs In-Store

Online | 6,354 txns

Online transactions (6,354) slightly exceeded In-Store transactions (6,221), indicating a nearly equal split between both channels with a marginal preference for online shopping.

4

Peak Sales Month

Jan 2022 | $52,911.50

January 2022 recorded the highest monthly revenue at $52,911.50. January 2025 had the lowest at $25,548.50 — however this is a partial month with data only up to January 18th.

5

Discount Impact

33.5% Orders | 4,219 orders

Out of 12,575 total transactions, 4,219 (33.5%) had discounts applied. Non-discounted orders (8,356) made up 66.5% of all sales, showing most purchases were at full price.

6

Credit Card Online

Online + CC | $260,851

Credit Card transactions made online generated $260,851, closely followed by Online 2024 sales of $265,246.50 — showing strong online purchasing behaviour in recent years.

7	**Discounted Beverages** **Beverages | $67,696** Discounted Beverages sales alone generated $67,696, highlighting that discount promotions in the Beverages category have a significant positive impact on revenue.

10. WORKBOOK SHEET GUIDE

SHEET NAME	PURPOSE
Raw_Data	Original untouched dataset — 12,575 rows, 11 columns. Never modified throughout the project.
Cleaned_Data	Cleaned and transformed data — all blanks fixed, 4 helper columns added. Total 15 columns.
Data_Analysis	All formulas and functions — SUM, AVERAGE, COUNTA, COUNTIF, SUMIF, SUMIFS, XLOOKUP, UNIQUE, SORT, FILTER, LARGE, IF Nested IF and Dynamic Arrays.
Pivot_Summary	4 Pivot Tables with calculated fields, 3 pivot charts, Category/Location/Payment Method slicers and Transaction Date timeline.
Advanced_Charts	Advanced charts — Combo Chart (Sales vs Order Count), Sparklines (trend indicators), Waterfall Chart (monthly revenue flow).
Dashboard	Final interactive KPI Dashboard — 6 KPI cards, 3 pivot charts, 3 slicers and 1 timeline all connected for real-time filtering.
Documentation	Full project documentation inside the workbook — cleaning steps, formulas used, sheet guide and key findings.

11. CONCLUSION

This Advanced Excel Mini Project successfully demonstrated the complete data analysis workflow using Microsoft Excel on a real-world Retail Store Sales dataset of 12,575 records spanning 3 years (2022 to 2025). The project covered all key areas as required by the Agileology course case scenario — from raw data cleaning to an interactive KPI dashboard.

The project validated that MS Excel (Advanced) is a powerful and industry-relevant tool for data analysis, capable of handling large datasets, building dynamic reports and delivering actionable business insights without the need for specialized BI tools.

SKILLS DEMONSTRATED	TOOLS & FUNCTIONS USED
Data Cleaning & Transformation	Go To Special, Remove Duplicates, Table Conversion
Advanced Formulas	SUMIFS, XLOOKUP, UNIQUE, SORT, FILTER, LARGE, Nested IF, TEXT, YEAR
Pivot Analysis	4 Pivot Tables, Calculated Field, Slicers, Timeline
Data Visualization	Column, Line, Pie, Combo, Sparklines, Waterfall Charts
What-If Analysis	Goal Seek, Scenario Manager
Dashboard Design	KPI Cards, Linked Charts, Interactive Filters
Documentation	Structured workbook with dedicated Documentation sheet